

Code-Layout, Readability and Reusability

Date	07 July 2026
Team ID	SWTID-2026-4350
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing the code /tabs used throughout the code	Yes	Proper indentation is maintained throughout the Flask application and model definitions.
2	Proper File Structure	Files and folders are logically organized	Partial	Main application is organized, but supporting files (config.py, prediction.py, templates) are stored separately
3	Meaningful Variable Name	Variables reflect their purpose clearly	Yes	Variables such as ApplicantDetail, ApprovalPrediction, CreditHistory, and predictor clearly describe their purpose
4	Function / Method Names	Functions are descriptively named	Yes	Functions like initialize_database() and get_predictor() use descriptive names.
5	Code Comments	Inline and block comments explain logic	Partial	Logging and readable code improve understanding, though additional inline comments can enhance maintainability.
6	Modular Design	Code is split into reusable functions/ modules	Yes	Database models, prediction service, configuration, and web routes are separated into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code avoids unnecessary duplication by using reusable database models and prediction functions.
8	Error Handling	Exceptions and errors are handled gracefull	Partial	Logging is configured and exceptions are partially handled. More comprehensive exception handling can be added for database and prediction failure

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Main web application and route handling	High
2	SQLAlchemy Models	Python, Flask-SQLAlchemy	Database operations throughout the application	High
3	Credit Predictor (CreditPredictor)	Python, Machine Learning	Loan approval prediction service	High
4	Database Initialization	Python	Database setup and default data seeding	Medium
5	HTML Templates (home.html, index.html, result.html)	HTML, Jinja2	User interface rendering	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured Flask application with clear separation of routes, models, and prediction logic.
Readability	4	Meaningful naming conventions and organized code improve readability. Additional comments can further enhance clarity.
Reusability	5	Modular architecture allows prediction logic, database models, and templates to be reused in future projects.
Documentation / Comments	4	Logging is available, but more detailed code comments and documentation would improve maintainability.
Overall Score	4.5	The project demonstrates good coding practices with modular design, reusable components, and a maintainable structure.

